Prompt Engineering & Experimentation

Date	13 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)

Prompt Engineering & Experimentation

Applicability Note

VBCUA does not use a prompt-driven generative language model (such as Gemini, GPT, or a similar chat/completion model). It uses two task-specific, pre-trained models — OpenAI Whisper for speech-to-text transcription and a Sentence Transformer for semantic similarity — both of which are invoked through direct function calls (transcribe(), encode()) rather than natural-language prompts. Consequently, most of the classic prompt-engineering activities below do not apply to the current implementation. Each point is addressed honestly below, drawing the closest real analogue in VBCUA's design where one exists, and marking items as Not Applicable where no equivalent exists.

Prompt Engineering & Experimentation	**Initial prompt design:** ***[Not Applicable]*** No natural-language prompt is used to elicit output from either model. Whisper transcribes audio directly, and the Sentence Transformer encodes text directly into embeddings; there is no prompt text to design. **System instructions / role definition:** There is no system prompt or assigned persona/role for a generative model. The closest functional equivalent is the fixed grading rubric implemented in concept_scorer.py — the score thresholds (90% / 75% / 60%) and their associated feedback text — which deterministically defines how the system's output is framed, in the same way a system instruction might guide an LLM's tone or behaviour. **Few-shot / zero-shot prompts:** ***[Not Applicable]*** No example-based prompting technique is used. The single stored reference answer functions as a fixed ground-truth comparison point for every submission, which is conceptually closer to a static evaluation rubric than to a few-shot or zero-shot prompt. **Prompt testing iterations:** As no prompts exist, no prompt-iteration process was carried out. The closest equivalent tuning activity in VBCUA is the selection of the score thresholds and feedback wording in concept_scorer.py, which are currently fixed initial values rather than the result of a documented calibration process. Future enhancement: running the pipeline against a labelled set of sample student responses to empirically tune the grade thresholds and feedback wording is identified as a valuable follow-up activity, but is not part of the current implementation. **Output evaluation criteria:** While there is no generated text to evaluate for quality in the LLM sense, VBCUA does define clear evaluation criteria for its own output: the cosine similarity score is converted into a percentage, and a grade tier (Excellent / Good / Fair / Needs Improvement) is assigned based on fixed thresholds, which serves as the system's output evaluation logic. **Prompt optimization:** ***[Not Applicable]*** Not applicable, as no prompt exists to optimise. **Safety & bias handling:**

	No explicit safety filtering or bias-mitigation logic is implemented in the current version. Automatic speech recognition models such as Whisper are known from published research to exhibit varying accuracy across accents, dialects, and speech patterns, and this has not been evaluated or addressed for VBCUA. Future enhancement: evaluating transcription and scoring accuracy across a range of accents and speech patterns, and documenting any observed disparities, is identified as a necessary step before the tool is used in a way that could affect a student's grade. **Final optimized prompt:** ***[Not Applicable]*** Not applicable, as no prompt is used in the current implementation.